

Nyström Method for 2D Electromagnetic Scattering (TM) EFIE + Nyström Discretization

Fluid Gravity

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Electric Field Integral Equation (EFIE) — TM Polarization

TM polarization: the fields and current have only a z-component:

$$\mathbf{E} = \hat{z}E_z, \quad \mathbf{J} = \hat{z}J.$$

For a 2D PEC scatterer, the EFIE on the boundary Γ reads

$$E_z^{\text{inc}}(\mathbf{r}) + j\omega\mu \int_{\Gamma} G(\mathbf{r}, \mathbf{r}') J(\mathbf{r}') ds' = 0,$$

with the 2D Green's function

$$G(\mathbf{r}, \mathbf{r}') = \frac{j}{4} H_0^{(2)}(k|\mathbf{r} - \mathbf{r}'|).$$

- Γ : boundary of the 2D PEC object
- $\mathbf{r}$: observation point, $\mathbf{r}'$: source point
- $H_0^{(2)}$: Hankel function of order 0, second kind
- ω : angular frequency, k : wavenumber, μ : permeability
- Unknown: surface current density $J(\mathbf{r}')$

Boundary Parametrization and EFIE Kernel

We parametrize the boundary Γ by a 2π -periodic curve

$$\mathbf{r}(t) = (x(t), y(t)), \quad 0 \leq t < 2\pi,$$

with tangent

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt}(t), \quad |\mathbf{r}'(t)| = \sqrt{x'(t)^2 + y'(t)^2}.$$

The EFIE becomes

$$E_z^{\text{inc}}(t) + j\omega\mu \int_0^{2\pi} M(t, \tau) J(\tau) d\tau = 0,$$

where the single-layer kernel is

$$M(t, \tau) = -\frac{j}{2} H_0^{(2)}(k|\mathbf{r}(t) - \mathbf{r}(\tau)|) |\mathbf{r}'(\tau)|.$$

The kernel is weakly singular as $t \rightarrow \tau$.

Regularization of the Singular Kernel

To handle the logarithmic singularity, we use the Martensen–Kussmaul decomposition (see Colton & Kress, *Inverse Acoustic and Electromagnetic Scattering Theory*):

$$M(t, \tau) = M_1(t, \tau) \ln \left(4 \sin^2 \frac{t - \tau}{2} \right) + M_2(t, \tau),$$

with

$$M_1(t, \tau) = -\frac{1}{2\pi} J_0(k|\mathbf{r}(t) - \mathbf{r}(\tau)|) |\mathbf{r}'(\tau)|.$$

The diagonal limit of the smooth part M_2 is

$$M_2(t, t) = \left(-\frac{j}{2} - \frac{\gamma}{\pi} - \frac{1}{\pi} \ln \left(\frac{k}{2} |\mathbf{r}'(t)| \right) \right) |\mathbf{r}'(t)|,$$

where γ is the Euler–Mascheroni constant.

For $t \neq \tau$,

Nyström Method — Boundary Discretization

We choose $2N$ equispaced midpoint nodes on $[0, 2\pi)$:

$$t_n = \frac{(2n+1)\pi}{2N}, \quad n = 0, 1, \dots, 2N-1.$$

At each node:

$$\mathbf{r}_n = \mathbf{r}(t_n), \quad \mathbf{r}'_n = \mathbf{r}'(t_n), \quad ds_n = |\mathbf{r}'_n|.$$

The unknown current is sampled as

$$J_n \approx J(t_n), \quad n = 0, \dots, 2N-1.$$

The integral is approximated by a trigonometric Nyström rule that treats the logarithmic singularity analytically via the decomposition

$$M = M_1 \ln(\cdot) + M_2.$$

Idea: use the same nodes and arc-length weights both for EFIE and far-field.

Colton–Kress Trigonometric Weights

The periodic logarithmic kernel

$$\ln\left(4 \sin^2 \frac{t - \tau}{2}\right)$$

is integrated using special trigonometric weights $R^{(N)}(t_i, t_j)$.

For the midpoint nodes t_i, t_j :

$$R^{(N)}(t_i, t_j) = -\frac{2\pi}{N} \sum_{m=1}^{N-1} \frac{1}{m} \cos(m(t_i - t_j)) - \frac{\pi}{N^2} \cos(N(t_i - t_j)).$$

In the discrete EFIE, these weights multiply $M_1(t_i, t_j)$, while $M_2(t_i, t_j)$ is integrated with the standard trapezoidal weight π/N .

Nyström Method — Discrete EFIE System

At each collocation node t_i we enforce the EFIE:

$$E_z^{\text{inc}}(t_i) + j\omega\mu \sum_{j=0}^{2N-1} \left[R^{(N)}(t_i, t_j) M_1(t_i, t_j) + \frac{\pi}{N} M_2(t_i, t_j) \right] J_j = 0.$$

This yields a linear system

$$\mathbf{A} \mathbf{J} = \mathbf{g},$$

with

$$A_{ij} = j\omega\mu \left[R^{(N)}(t_i, t_j) M_1(t_i, t_j) + \frac{\pi}{N} M_2(t_i, t_j) \right],$$

and

$$g_i = -E_z^{\text{inc}}(t_i).$$

In the MATLAB code, $\mathbf{A}$ is assembled exactly in this form and solved via $\mathbf{J} = \mathbf{A}^{-1} \mathbf{g}$.

Circular Cylinder and Reference Current (Balanis)

For a circular PEC cylinder of radius a and a plane wave

$$E_z^{\text{inc}} = E_0 e^{-jkx},$$

the exact surface current on $\rho = a$ can be written as a cylindrical series (Balanis):

$$J_s(\phi) = \hat{z} \frac{2E_0}{\pi a \omega \mu} \sum_{n=-\infty}^{+\infty} j^{-n} \frac{e^{jn\phi}}{H_n^{(2)}(ka)}.$$

In the code, this is implemented as

$$J_{\text{ref}}(\phi) = \frac{2E_0}{\pi a k \eta_0} \sum_{n=-N_{\text{max}}}^{N_{\text{max}}} \frac{j^{-n} e^{jn\phi}}{H_n^{(2)}(ka)},$$

with $k\eta_0 = \omega\mu$ and a finite truncation N_{max} .

The numerical Nyström current $J(t_n)$ is compared against $J_{\text{ref}}(t_n)$ in both magnitude and phase.

Far-Field Pattern from Nyström Current

In the far field ($\rho \rightarrow \infty$), the scattered field behaves as

$$E_z^{\text{scat}}(\rho, \phi) \sim \sqrt{\frac{2}{\pi k \rho}} e^{-j(k\rho - \pi/4)} F(\phi),$$

where the angular far-field pattern is

$$F(\phi) = \frac{j\omega\mu}{4} \int_{\Gamma} e^{jk\hat{\mathbf{r}} \cdot \mathbf{r}'} J(\mathbf{r}') ds', \quad \hat{\mathbf{r}} = (\cos \phi, \sin \phi).$$

Nyström approximation:

$$F(\phi) \approx \frac{j\omega\mu}{4} \sum_{n=0}^{2N-1} e^{jk(X_n \cos \phi + Y_n \sin \phi)} J_n ds_n.$$

For a circular cylinder, the reference far field is given by the Mie series

$$F_{\text{ref}}(\phi) = -E_0 \sqrt{\frac{2j}{\pi k}} \sum_{n=-\infty}^{+\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{jn\phi}.$$

Summary

- EFIE (TM) for 2D PEC scattering is formulated with a single-layer operator.
- The kernel is weakly singular and is regularized via the Martensen–Kussmaul decomposition $M = M_1 \ln(\cdot) + M_2$.
- Colton–Kress trigonometric weights provide a high-order Nyström quadrature for the periodic logarithmic kernel.
- The resulting linear system is assembled and solved in MATLAB.
- The induced current and far-field pattern are validated against analytic cylindrical (Mie) series for a circular cylinder.

This framework extends naturally to smooth non-circular boundaries (e.g., ellipses) by changing the parametrization $\mathbf{r}(t)$.